

Explainable AI:

Enhancing Transparency and Trust

AI is giving us answers to a million questions, but we don't really know how it arrives at them. Explainable AI (XAI) helps understand what went into making a decision, enhancing trust in the system, especially with respect to ethical practices.



Artificial intelligence (AI) is transforming industries with sophisticated decision-making and automation technologies. However, as AI systems become more complicated, the need for openness and interpretability to understand and trust AI-driven choices grows. The field of explainable AI (XAI) has emerged just to address this.

Explainable AI and model interpretability are critical not just for comprehending AI decisions, but also for fostering trust and assuring ethical AI practices. As AI continues to enter numerous industries, the demand for transparent and interpretable AI models grows. Striking the right balance between transparency and performance,

using relevant XAI methodologies, and adhering to legislation and ethical principles are crucial for realising AI's full potential in a responsible and accountable manner.

As AI plays an increasingly important role in our lives, it is critical that AI systems are transparent, accountable, and adhere to ethical norms. Adopting explainable AI not only assures compliance with present and new rules, but also upholds ethical norms, allowing AI to benefit society while respecting individual rights and values.

Table 1 gives a quick overview of how explainable AI fits into the paradigm of artificial intelligence research activities.

The need for explainable AI

The field of AI has experienced exponential growth, driven by factors such as the increasing volume of data, post-COVID expectations in automation, and the pursuit of operational excellence in digital experiences. There are multiple factors that have contributed to the importance of explainable AI.

Trust and accountability:

AI decisions in vital fields like healthcare, banking, and autonomous cars can have life-changing effects. Understanding how and why AI arrives at certain conclusions is critical for building trust and responsibility.